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Hot Flashes and Night Sweats (PDQ®)

Health Professional Version

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PDQ Supportive and Palliative Care Editorial Board.

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This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the pathophysiology and treatment of hot flashes and night sweats. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

This summary is reviewed regularly and updated as necessary by the PDQ Supportive and Palliative Care Editorial Board, which is editorially independent of the National Cancer Institute (NCI). The summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

Overview

Hot flashes and night sweats are common in cancer survivors, particularly women, but they can also occur in men. Pathophysiological mechanisms are complex. Treatment options are broad-based and include hormonal agents, nonhormonal pharmacotherapies, and diverse integrative medicine modalities.[1]

Hot flashes, also called hot flushes, occur in approximately two-thirds of postmenopausal women with a history of breast cancer and are associated with night sweats in 44% of these women.[2,3] The severity of hot flashes in patients with breast cancer has been associated with sleep difficulty, higher pain severity, and poor psychological functioning.[4] In premenopausal breast cancer survivors, vasomotor symptoms—including hot flashes and night sweats—have been associated with depression, an effect that may be mediated by sleep disturbance.[5] For most patients with breast cancer or prostate cancer, hot flash intensity is moderate to severe. Sweating can be part of the hot flash complex that characterizes the vasomotor instability of menopause. Physiologically, sweating mediates core body temperature by producing transdermal evaporative heat loss.[6,7] Hot flashes accompanied by sweating that occur during the sleeping hours are often called **night sweats**.[8]

Approximately 20% of women without breast cancer seek medical treatment for postmenopausal symptoms, including symptoms related to vasomotor instability.[9] Vasomotor symptoms resolve spontaneously in most patients in this population, with only 20% of affected women reporting significant hot flashes 4 years after the last menses.[9] There are no comparable data for women with metastatic breast cancer. Three-quarters of men with locally advanced or metastatic prostate cancer treated with medical or surgical orchiectomy experience hot flashes.[10]

In this summary, unless otherwise stated, evidence and practice issues as they relate to adults are discussed. The evidence and application to practice related to children may differ significantly from information related to adults. When specific information about the care of children is available, it is summarized under its own heading.

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Etiology

Causes of menopausal hot flashes include natural menopause, surgical menopause, or chemical menopause. In patients with cancer, chemical menopause may be caused by the following:

- Cytotoxic chemotherapy.
- Radiation therapy.
- Androgen treatment.

Causes of so-called male menopause include the following:

- Orchiectomy.
- Gonadotropin-releasing hormone use.
- Estrogen use.

Drug-associated causes of hot flashes and night sweats in men and women include use of the following:

- Tamoxifen.
- Aromatase inhibitors.
- Opioids.
- Tricyclic antidepressants.
- Steroids.

Women who are extensive metabolizers of tamoxifen related to *CYP2D6* may have more severe hot flashes than women who are poor metabolizers;^[1] however, conflicting data surround this topic.^[2]

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Primary Interventions

Hormone Replacement Therapy

Estrogen replacement effectively controls hot flashes associated with biological or treatment-related postmenopausal states in women. The proposed mechanism of action of estrogen replacement therapy is that it ameliorates hot flashes by raising the core body temperature sweating threshold.^[1] [Level of evidence: I] However, many women have relative or absolute contraindications to estrogen replacement. Physicians and breast cancer survivors often think there is an increased risk of breast cancer recurrence or de novo breast malignancy with hormone replacement therapies and defer hormonal management of postmenopausal symptoms. Methodologically strong data evaluating the risk of breast cancer associated with hormone replacement therapy in healthy women have been minimal, despite strong basic science considerations suggesting the possibility of such a risk.^[2]

The Women's Health Initiative, a large, randomized, placebo-controlled trial of the risks and benefits of estrogen plus progestin in healthy postmenopausal women, was stopped prematurely in 2002 at a mean follow-up of 5.2 years (± 1.3) because researchers found a 1.26-fold increased risk of breast cancer (95% confidence interval [CI], 1.00–1.59) in women receiving hormone replacement therapy. Tumors among women in the hormone replacement therapy group were slightly larger and more advanced than tumors among women in the placebo group, with a substantial and statistically significant rise in the percentage of abnormal mammograms at first annual screening. This rise might hinder breast cancer diagnosis and account for the later stage at diagnosis.^[3,4] [Level of evidence: I] These results are

supported by a population-based case-control study suggesting a 1.7-fold increased risk of breast cancer (95% CI, 1.3–2.2) in women using combined hormone replacement therapy. The risk of invasive lobular carcinoma was increased 2.7-fold (95% CI, 1.7–4.3), the risk of invasive ductal carcinoma was increased 1.5-fold (95% CI, 1.1–2.0), and the risk of estrogen receptor–positive/progesterone receptor–positive breast cancer was increased 2.0-fold (95% CI, 1.5–2.7). The increased risk was highest for invasive lobular tumors and in women who used hormone replacement therapy for longer periods. The risk was not increased with the use of unopposed estrogen therapy.[5]

The limited data do not indicate an increased risk of breast cancer recurrence with single-agent estrogen use in patients with a history of breast cancer.[6,7] A series of double-blind, placebo-controlled trials suggested that low-dose megestrol acetate is a promising agent for hot flash management in this population.[8][Level of evidence: I]; [9][Level of evidence: II] Limited data suggest that brief cycles of intramuscular depot medroxyprogesterone acetate also play a role in the management of hot flashes.[10][Level of evidence: I] The risk associated with progestin use is unknown.[2]

Examples of hormone-based pharmacological treatments for vasomotor symptoms are summarized in Table 1.

Table 1. Hormone-Based Treatment of Vasomotor Symptoms

Drug Category	Medication	Dose	Comment	Reference
Estrogen	Example: 17-beta-estradiol	0.5 mg PO q24h	Multiple routes available; consider use of estrogen/progestin combination products for women with intact uteri.	[11]; [12][Level of evidence: I]
Progestin	Megestrol acetate	20 mg PO qd	Studied in men and women.	[8][Level of evidence: I]; [9][Level of evidence: III]
	Medroxyprogesterone	400 mg IM x 1		[13][Level of evidence: I]; [14][Level of evidence: III]

IM = intramuscular; PO = by mouth; qd = every day.

Other Pharmacological Interventions

Numerous nonestrogenic pharmacological treatments for hot flash management in women with a history of breast cancer and in some men who have undergone androgen deprivation therapy have been evaluated. Options with reported efficacy include the following: [15–17][Level of evidence: I]

- Androgens.
- Progestational agents.
- Gabapentin.
- Selective serotonin reuptake inhibitors (SSRIs).
- Selective serotonin norepinephrine inhibitors.
- Alpha adrenergic agonists (e.g., methyldopa, clonidine).
- Beta-blockers.

Inferior efficacy, lack of large definitive studies, and potential side effects limit the use of many of these agents.

Agents that have been found to be helpful in large, randomized, placebo-controlled clinical trials include the following: [15–17]

- Venlafaxine.
- Paroxetine.
- Citalopram.
- Fluoxetine.
- Gabapentin.
- Pregabalin.

- Clonidine.

Agents that confer a 55% to 60% reduction in hot flashes are venlafaxine extended release,[18] paroxetine controlled release [19,20] [Level of evidence: I] or immediate release,[21] gabapentin,[22-25][Level of evidence: I]; [26][Level of evidence: II] and pregabalin.[27] [Level of evidence: I] Other effective agents resulting in a reduction in hot flashes of approximately 50% include citalopram [28][Level of evidence: I] and fluoxetine.[29][Level of evidence: I] Clonidine, transdermal [30] or oral,[31][Level of evidence: I] can reduce hot flashes by approximately 40%.

One study compared the efficacy and patient preference of venlafaxine, 75 mg once daily, to gabapentin, 300 mg 3 times daily, for the reduction of hot flashes. Sixty-six women with a history of breast cancer were randomly assigned in an open-label fashion to receive venlafaxine or gabapentin for 4 weeks. After a 2-week washout period, they received the opposite treatment for an additional 4 weeks. Both treatments reduced hot flash scores (severity multiplied by frequency) by approximately 66%. However, significantly more women preferred venlafaxine to gabapentin (68% vs. 32%, respectively).[22]

A study using citalopram examined how much of a reduction in hot flashes was needed to have a positive impact on activities of daily living and general health-related quality of life.[32] The authors reported that hot flashes had to be reduced by at least 46% for women to report significant improvements in the degree of bother they experienced in daily activities.

In a randomized study of paroxetine versus placebo in postmenopausal survivors of gynecological cancer, paroxetine significantly reduced the severity and frequency of hot flashes and nighttime awakening attributed to vasomotor symptoms, with improvement in sleep duration.[20][Level of evidence: I]

Agents that have been evaluated in phase II trials but have not shown efficacy include bupropion,[33] aprepitant,[34] and desipramine. [35][Level of evidence: II] Interestingly, these agents do not primarily modulate serotonin. In addition, randomized clinical trials with sertraline have not provided convincing evidence of its efficacy in hot flash management.[36-38][Level of evidence: I]

Examples of nonhormonal pharmacological treatments for vasomotor symptoms are summarized in Table 2.

Table 2. Nonhormonal Treatment of Vasomotor Symptoms

Drug Category	Medication	Dose	Comment	Reference
Selective serotonin reuptake inhibitor	Citalopram	10–20 mg PO q24h	Mixed efficacy results	[28][Level of evidence: I]
	Escitalopram	10–20 mg PO q24h	Studied in a nononcology patient population	[39,40][Level of evidence: I]
	Fluoxetine	20 mg PO q24h		[29][Level of evidence: I]
	Paroxetine	IR: 10–20 mg PO q24h CR: 12.5–25 mg PO q24h	Brisdelle branded product for VMS 7.5 mg PO qhs	[19-21][Level of evidence: I]
	Sertraline	50 mg PO q24h	Benefit seen over placebo after crossover, but not vs. baseline VMS	[36][Level of evidence: I]
Serotonin/norepinephrine reuptake inhibitor	Venlafaxine	37.5–150 mg/d (daily dosing for ER or in 2–3 divided doses for IR for doses >37.5 mg)		[11]; [18,41,42] [Level of evidence: I]
	Duloxetine	30 mg qam x 1 wk, then 60 mg qam	Equivalent to escitalopram (10 mg qam x 1 week, then 20 mg qam) in reducing hot flash severity and frequency and depressive symptoms	[43][Level of evidence: I]
Alpha-2 antagonist antidepressant	Mirtazapine	7.5–30 mg qhs	Small pilot trial; target dose, 15–30 mg	[44][Level of evidence: II]
Anticonvulsant/GABA analogue	Gabapentin	Initial, 300 mg qhs; titrate up to 900 mg/d in divided doses	Mixed results depending on comparator group; studied in men and women	[22-24,26] [Level of evidence: I]

Drug Category	Medication	Dose	Comment	Reference
	Pregabalin	50 mg qhs, then 50–150 mg PO bid	Titration should be made weekly, to a target dose of 75 mg PO bid	[11]; [27][Level of evidence: I]
Alpha-2 adrenergic agonist	Clonidine	0.1 mg/24 h transdermal; 0.1 mg PO q24h	Sudden cessation can result in significant hypertension; no efficacy demonstrated in men with postorchiectomy hot flashes	[31,45][Level of evidence: I]

bid = twice a day; CR = controlled release; ER = extended release; GABA = gamma-aminobutyric acid; IR = immediate release; PO = by mouth; qam = every morning; qhs = once daily at bedtime; VMS = vasomotor symptoms.

If nighttime hot flashes or night sweats are a particular problem without causing much bother during daytime, strategies to simultaneously improve sleep and hot flashes are in order. Limited data exist related to effective treatments that can target both symptoms. One pilot trial evaluated mirtazapine (a tetracyclic antidepressant that mainly affects serotonin) for hot flashes because it is often prescribed for sleep difficulties. Twenty-two women received mirtazapine titrated up to 30 mg per day at bedtime over a 3-week period; then they could choose 15 mg or 30 mg at bedtime daily for the fourth week. Hot flashes were reduced by approximately 53% in this nonrandomized trial, and women were statistically significantly satisfied with their hot flash control.[44] However, only 16 of the 22 women stayed on the agent for the entire study period because of excessive grogginess. Although this agent could be further studied in a larger randomized trial, it would be particularly important to evaluate the risk/benefit ratio.

In the short term, side effects of antidepressant agents in the doses used to treat hot flashes are minimal and primarily include:

- Nausea.
- Sedation.
- Dry mouth.
- Appetite suppression or stimulation.

In the long term, the prevalence of decreased sexual function with the use of SSRIs at doses for treating hot flashes is not known. The anticonvulsants gabapentin and pregabalin can cause sedation, dizziness, and difficulty concentrating, while clonidine can cause dry mouth, sedation, constipation, and insomnia.[25,27,46][Level of evidence: I] Patients respond as individuals to both the efficacy and the toxicity of various medications. Therefore, careful assessment and tailored treatment chosen collaboratively by the provider and patient are needed.

Data indicate that if a medication does not help an individual, switching to another medication—whether a different antidepressant or gabapentin—may be worthwhile. A randomized phase III trial (NCCTG-N03C5) examined gabapentin alone versus gabapentin with an antidepressant in women who had inadequate control of their hot flashes with an antidepressant alone.[47][Level of evidence: I] Gabapentin use resulted in an approximately 50% median reduction in hot flash frequency and score, regardless of whether the antidepressant was continued. In other words, for women who were using antidepressants exclusively to manage hot flashes that were inadequately controlled, initiation of gabapentin with discontinuation of the antidepressant produced results equal to those obtained with combined therapy, resulting in the need for fewer medications. Similarly, in a pilot study of women receiving inadequate benefit from venlafaxine for hot flash reduction, switching to open-label citalopram, 20 mg per day, resulted in a 50% reduction in hot flash frequency and score.[48]

Drug Interactions

Many SSRIs can inhibit the cytochrome P450 enzymes involved in the metabolism of tamoxifen, which is commonly used in the treatment of breast cancer. When SSRIs are used, drug-drug interactions are noted. Tamoxifen is metabolized by the cytochrome P450 enzyme system, specifically CYP2D6. Wild-type CYP2D6 metabolizes tamoxifen to an active metabolite, 4-hydroxy-N-desmethyl-tamoxifen, also known as endoxifen. A prospective trial evaluating the effects on tamoxifen metabolism of the coadministration of tamoxifen and paroxetine, a CYP2D6 inhibitor, found that paroxetine coadministration resulted in decreased concentrations of endoxifen. The magnitude of decrease was greater in women with the wild-type CYP2D6 genotype than in those with a variant genotype ($P = .03$). [49][Level of evidence: II]

In a prospective observational study of 80 women initiating adjuvant tamoxifen therapy for newly diagnosed breast cancer, variant CYP2D6 genotypes and concomitant use of SSRI CYP2D6 inhibitors resulted in reduced endoxifen levels. Variant CYP2D6 genotypes do not produce functional CYP2D6 enzymes.[50][Level of evidence: II] Since this study was published, several researchers have evaluated the clinical implications of this finding.[51];[52-54][Level of evidence: II] One study followed more than 1,300 women for a

median of 6.3 years and concluded that women who were poor metabolizers or heterozygous extensive/intermediate metabolizers (hence, less CYP2D6 activity) had higher rates of breast cancer recurrence, worse event-free survival, and worse disease-free survival than women who were extensive metabolizers.[53] Similarly, authors of a retrospective cohort study of more than 2,400 women in Ontario, Canada, who were being treated with tamoxifen and had overlapping treatment with an SSRI concluded that women who concomitantly used paroxetine and tamoxifen had an increased risk of death that was proportionate to the amount of time they used these agents together.[54][Level of evidence: II]

The clinical implications of these changes and of other CYP2D6 genotypes [55] have not yet been elucidated, but the pharmacokinetic interaction between tamoxifen and the newer antidepressants used to treat hot flashes merits further study.[56] Likewise, the risk of soy phytoestrogen use on breast cancer recurrence and/or progression has not yet been clarified. Soy phytoestrogens are weak estrogens found in plant foods. *In vitro* models suggest that these compounds have a biphasic effect on mammary cell proliferation that is dependent on intracellular concentrations of phytoestrogen and estradiol.[57]

Information Specific to Men

Data are scant regarding the pathophysiology and management of hot flashes in men with prostate cancer. The rate of hot flashes in men receiving androgen deprivation therapy is approximately 75%.[58] The limited data suggest that hot flashes in men are related to changes in sex hormone levels that cause instability in the hypothalamic thermoregulatory center. This is analogous to the proposed mechanism of hot flashes that occur in women. As in women with breast cancer, hot flashes impair the quality of life for men with prostate cancer who are receiving androgen deprivation therapy. The vasodilatory neuropeptide, calcitonin gene-related peptide, may be instrumental in the genesis of hot flashes.[58]

In a prespecified secondary analysis of a prostate cancer clinical trial, 93% of men receiving 12 months of androgen deprivation therapy experienced hot flashes. The hot flashes occurred at castrate levels of testosterone, and cessation of hot flashes preceded full recovery of testosterone in most men, with 99% of men reporting resolution of hot flashes.[59][Level of evidence: I]

Cognitive behavioral therapy (CBT) has been studied for the treatment of hot flashes in men undergoing androgen deprivation therapy for prostate cancer.[60][Level of evidence: I] Patients were randomly assigned to a guided self-help CBT regimen, which included a booklet and CD with relaxation and breathing exercises, or to treatment as usual. At 6 weeks, those assigned to CBT experienced a statistically significant 40% reduction in hot flash/night sweat symptoms versus a 12% reduction in patients who received treatment as usual. Symptom reduction continued but was not statistically significant at 32 weeks. Adherence to CBT was good, with 88% reading all or more than half of the booklet and 79% using the relaxation CD.

With the exception of clonidine, the agents used to treat hot flashes in women have shown similar efficacy in men. For more information, see the [Other Pharmacological Interventions](#) section. Treatment modalities for men have included the following:[61]

- Estrogens.
- Progesterone.
- SSRIs.
- Gabapentin.

One large, multisite study from France [62] randomly assigned men who were taking leuporelin for prostate cancer to receive venlafaxine, 75 mg; cyproterone acetate (an antiandrogen), 100 mg; or medroxyprogesterone acetate, 20 mg, when they reported at least 14 hot flashes per week. All three treatments significantly reduced hot flashes at 8 weeks, with cyproterone acetate resulting in a 100% median reduction, medroxyprogesterone resulting in a 97% reduction, and venlafaxine resulting in a 57% reduction. More adverse events were reported with cyproterone acetate, including one serious adverse event (dyspnea) attributable to the drug. Venlafaxine was not associated with any serious adverse events and overall had a 20% adverse event rate. Medroxyprogesterone was the best-tolerated drug, with an adverse event rate of 12%, but with one serious event, urticaria. The most frequent side effects for all agents were gastrointestinal issues, including nausea, constipation, diarrhea, and abdominal pain.[62]

Based on its efficacy in women, the combination of venlafaxine and soy was studied for hot flash reduction in androgen-deprived men. [63][Level of evidence: I] Patients were randomly assigned to receive venlafaxine with soy protein, venlafaxine with milk protein placebo, soy protein with placebo, or dual placebos during a 12-week period. The number and severity of hot flashes fell for all arms during the study period, but there was no significant difference between the arms. The authors concluded that neither agent should be used to treat hot flashes in men, that there is a significant placebo effect in the study of hot flash treatment, and that agents demonstrating success for hot flashes in women may not be successful in men.

A small, multicenter, retrospective review evaluated the use of two doses of intramuscular medroxyprogesterone acetate (400 mg and 150 mg) as a single dose to treat and prevent hot flashes associated with luteinizing hormone-releasing hormone agonist therapy for prostate cancer.[14][Level of evidence: III] Of the 48 men studied, 91% experienced symptomatic improvement in hot flashes, and 46% experienced complete resolution of hot flashes. The trial was not powered to detect a difference between the two doses; however, the authors concluded that they would use the 400-mg dose.

Pilot studies of the efficacy of the SSRIs paroxetine and fluvoxamine suggest that these drugs decrease the frequency and severity of hot flashes in men with prostate cancer.[64,65] As in women with hormonally sensitive tumors, there are concerns about the effects of hormone use on the outcome of prostate cancer, in addition to other well-described side effects.[58]

Cognitive and Behavioral Methods

Comprehensive nonpharmacological interventions have been developed and evaluated for their ability to reduce hot flashes, night sweats, and the related perception of burden or problems. These interventions have typically included the following:[66-69]

- Psychoeducation about managing general symptoms, including stress, anxiety, and sexual and other menopausal concerns.
- Relaxation exercises, including slow, deep breathing, called **paced breathing**.
- Cognitive restructuring that addresses catastrophizing, negative beliefs, and avoidance behaviors.

Behavioral interventions as a primary or adjunctive modality may also play a role in hot flash management. Core body temperature has been shown to increase before a hot flash;[70] therefore, interventions that control body temperature could improve hot flash management. Some methods of controlling body temperature include the use of the following:

- Loose-fitting cotton clothing.
- Fans and open windows to keep air circulating.

Since serotonin may be involved as a central hot flash trigger, behavioral interventions such as stress management may modulate serotonin, causing a decrease in hot flashes.

Relaxation training and paced breathing were initially found to decrease hot flash intensity by as much as 40% to 50% in controlled pilot trials;[71,72] however, randomized trials with control arms using a different pace of breathing have not demonstrated significant benefit for paced-breathing interventions.[73,74]

Three large studies [67-69] of similar interventions have been completed using no treatment, usual care, or wait-list control comparison groups. While all of the studies demonstrated significant reductions in problem ratings or bother ratings related to hot flashes and night sweats, none showed actual reductions in hot flash frequency. Only one of the three studies demonstrated some significant improvements in night sweats at some data points.[68] Similar results were seen in a large trial of Internet-based CBT with and without therapist support.[75][Level of evidence: I] Cognitive behavioral interventions may be an important addition to pharmacological treatment to improve a patient's symptoms related to hot flashes. However, data have not supported the sole use of CBT for reducing hot flashes.

Medical hypnosis is a newer intervention for hot flashes that has been shown to be helpful. In medical hypnosis, the provider facilitates a deep relaxation and trance state in the patient and gives suggestions to the subconscious to mitigate the symptom or problem being addressed. For hot flashes, medical hypnosis uses cooling suggestions and stress reduction to prevent rises in core body temperature and to decrease sympathetic activation. Based on strong pilot data, a randomized controlled trial of 187 postmenopausal women used an attention-control comparison and demonstrated significantly greater reductions in hot flashes in the hypnosis group than in the control group. The hypnosis intervention was 5 weeks long. At week 6, hot flash frequency was reduced in the hypnosis group by 64%, compared with a 9% reduction in the control group. At week 12, the reduction in the hypnosis group was 75%, compared with a 17% reduction in the control group.[76] Cancer survivors were not included in this study, but previous research has not demonstrated that interventions have a differential effect on hot flashes on the basis of breast cancer history.

Future research on hot flash management may be aided by the development of psychometrically sound assessment tools such as the Hot Flash Related Daily Interference Scale, which evaluates the impact of hot flashes on a wide variety of daily activities.[77]

Integrative Approaches

Herbs and dietary supplements

Numerous herbs and dietary supplements are popularly used for hot flash reduction. Several of these substances have not been well studied in rigorous clinical trials. Furthermore, the biological activity of various over-the-counter supplements has yet to be determined

and is far from standardized. Some of the more well-studied agents include vitamin E, soy phytoestrogen, and black cohosh.

Vitamin E, 400 IU twice a day, appears to confer a modest reduction in hot flashes that is only slightly better than that seen with placebo. The reduction in hot flashes is roughly 35% to 40%.^[78,79][Level of evidence: I]

Soy is a dietary supplement of interest for decreasing menopausal symptoms and breast cancer. The interest comes primarily from association studies of a high-soy diet and decreased breast cancer/menopausal symptoms in Asia. Soy is rich in isoflavones, which are part of a much larger class of plant compounds called flavonoids. Among the isoflavones in soy products are three compounds that are responsible for hormonal effects: genistein, daidzein, and glycitein.

Isoflavones are often referred to as phytoestrogens or plant-based estrogens because they have been shown, in cell line and animal studies, to bind with the estrogen receptor.^[80]

There is some confusion about the mechanisms of action and safety of these plant-based estrogens. Isoflavones have properties that can cause estrogen-like effects in some cells, for example, causing them to proliferate (divide and grow). But in other cells, isoflavones can stop or block estrogen effects, causing unwanted cells to not grow or die. There is continuing debate about the following questions:^[81]

- What doses and types of soy products inhibit estrogen as a growth factor?
- Under what circumstances do soy products inhibit estrogen as a growth factor?
- In what doses or circumstances do soy products promote estrogen-related growth?

Definitive answers to these questions are not known, but phytoestrogens continue to be investigated for their chemopreventive properties. On the other hand, soy has been well studied in numerous randomized, placebo-controlled trials for its effects on reducing hot flashes.^[82-86][Level of evidence: I] Most trials show that soy is no better than placebo in reducing hot flashes.^[87][Level of evidence: I];^[88] While clinical evidence indicates the general safety of soy products, there are no compelling data that would inspire the use of soy for hot flash management.

Similarly, randomized placebo-controlled trials of black cohosh have found that it is no better than a placebo in reducing hot flashes.^[86,89,90][Level of evidence: I] Furthermore, a meta-analysis that included 14 randomized controlled trials of black cohosh concluded there is a lack of evidence to support its use in the treatment of hot flashes.^[91] For more information, see [Black Cohosh](#).

Black cohosh used to be thought of as having estrogenic properties, but it is now known to act on serotonin receptors. One study evaluated black cohosh, red clover, estrogen and progesterone, and placebo in a randomized double-blind trial.^[92][Level of evidence: I] Each treatment arm was small (n = 22); however, over 12 months, hot flashes were reduced 34% by black cohosh, 57% by red clover, 63% by placebo, and 94% by hormone therapy. Of note, adherence rates were approximately 89% across the four groups during this long-term study. At 12 months, physiological markers such as endometrial thickness, estradiol, estrone, follicle-stimulating hormone, sex hormone-binding globulin, and liver function were not statistically different for those who took either red clover or black cohosh, compared with those who took a placebo. However, because these groups were small, the power for this secondary analysis was not reported, and it was likely underpowered to detect important differences.

Flaxseed is a plant that is part of the genus *Linum*, native to the area around the eastern Mediterranean and India. Flaxseed is a rich source of fiber, lignans, and omega-3 fatty acids. Lignans found in flaxseed are secoisolariciresinol diglucoside and alpha-linolenic acid. Lignans are a type of phytoestrogen (plant estrogen) that, like soy, are thought to have estrogen agonist-antagonist effects as well as antioxidant properties. Lignans are converted by colonic bacteria to enterodiol and enterolactone, which are metabolites believed to have important physiological properties, such as decreasing cell proliferation and inhibiting aromatase, 5-alpha reductase, and 17-beta hydroxysteroid activity. Cell line studies have shown properties of aromatase inhibition with enterolactone but less so with enterodiol.^[93] It is thought that these properties can reduce the risk of hormone-sensitive cancers.^[94-96] In addition, studies have shown that flaxseed can reduce estrogen levels through excretion in the urine.^[97,98]

Following preliminary test results of flaxseed for its effect on hot flashes and related endpoints,^[99,100][Level of evidence: I] an open-label pilot study evaluated 40 g of flaxseed in decreasing hot flashes. This study of 30 women showed a 57% reduction in hot flash scores and a 50% reduction in hot flash frequency over a 6-week period.^[101] However, a follow-up phase III, randomized, controlled trial conducted by the North Central Cancer Treatment Group with 188 women failed to show any benefit of 410 mg of lignans in a flaxseed bar over placebo.^[102][Level of evidence: I]

Similarly, on the basis of two pilot studies suggesting that magnesium oxide supplementation significantly reduced hot flashes, a double-blind, randomized, placebo-controlled trial of magnesium oxide, 800 or 1,200 mg daily, versus placebo was conducted in postmenopausal women with a history of breast cancer and symptomatic hot flashes.^[103][Level of evidence: I] No benefit was observed for magnesium oxide.

Many other plants and natural products are advertised or marketed as remedies for hot flashes. Some of these products, such as red clover, contain phytoestrogens, and some have unknown properties. The agents include dong quai, milk thistle, licorice, and chaste tree berry. There is incomplete understanding of the biology of these agents and whether taking them would impact breast cancer risk or recurrence in a negative or positive way. Data suggest that these plants have different effects, dependent not only on the dose used but also on a woman's hormone environment when she takes them. Little is known about these agents, and caution is needed with respect to taking them if a woman is to avoid estrogen supplementation.[104-107]

Acupuncture

Pilot and randomized sham trials have evaluated the use of acupuncture to treat hot flashes.[108-112][Level of evidence: I] Research in acupuncture is difficult to conduct, owing to the lack of novel methodology—specifically, the conundrum of what serves as an adequate control arm. In addition, the philosophy surrounding acupuncture practice is quite individualized, in that two women experiencing hot flashes would not necessarily receive the same treatment. It would be important to study acupuncture utilizing relevant clinical procedures; so far, acceptable research methods to accomplish this are lacking. Therefore, the data on the effect of acupuncture on hot flashes are quite mixed. However, a 2016 meta-analysis of 12 trials studying acupuncture for the treatment of hot flashes in patients with breast cancer showed limited or no effects.[113] Included trials ranged in size from 10 to 84 patients, with 5 to 16 treatment sessions and 1 to 24 months of follow-up. Comparator arms included hormone therapy, relaxation techniques, sham acupuncture, and antidepressants. The authors concluded that acupuncture failed to demonstrate a significant effect on the frequency of hot flashes in a population of breast cancer patients.

In contrast, a randomized controlled trial that was not included in the 2016 meta-analysis showed a statistically significant reduction in hot flash score with acupuncture.[114][Level of evidence: I] The trial randomly assigned women to ten acupuncture sessions plus enhanced self-care versus enhanced self-care alone. Women were included if they had breast cancer; at least moderate-level hot flashes, defined as six or more hot flashes a day; and/or a score of 15 or higher on the Greene Climacteric Scale. Random assignment to acupuncture resulted in fewer hot flashes and higher quality of life. The reduction in hot flash score was maintained through the 3- and 6-month follow-up visits.

In a randomized controlled trial, breast cancer survivors with hot flashes (120 women) were randomly assigned to receive electroacupuncture (using a transcutaneous electrical nerve stimulation unit to induce a current between two acupuncture points) or gabapentin, 900 mg daily, with sham electroacupuncture (needles that did not penetrate the skin and without electricity) and placebo capsules as controls.[24] Electroacupuncture produced the greatest reduction in hot flash symptoms, followed by sham acupuncture, gabapentin, and placebo capsules. In a separately published, prespecified secondary analysis of sleep outcomes in women assigned to the active treatment arms, electroacupuncture was comparable to gabapentin for improving sleep quality; significant associations were seen between reduction in hot flash severity and frequency and improved sleep latency and sleep quality in the full sample.[115]

For more information, see the [Vasomotor symptoms](#) section in [Acupuncture](#).

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Latest Updates to This Summary (01/15/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Editorial changes were made to this summary.

This summary is written and maintained by the PDQ Supportive and Palliative Care Editorial Board, which is editorially independent of NCI. The summary reflects an independent review of the literature and does not represent a policy statement of NCI or NIH. More information about summary policies and the role of the PDQ Editorial Boards in maintaining the PDQ summaries can be found on the [About This PDQ Summary](#) and [PDQ® Cancer Information for Health Professionals](#) pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the pathophysiology and treatment of hot flashes and night sweats. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

This summary is reviewed regularly and updated as necessary by the PDQ Supportive and Palliative Care Editorial Board, which is editorially independent of the National Cancer Institute (NCI). The summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

The lead reviewers for Hot Flashes and Night Sweats are:

- Larry D. Cripe, MD (Indiana University School of Medicine)
- Edward B. Perry, MD (VA Connecticut Healthcare System)
- Rachel A. Pozzar, PhD, RN (Dana-Farber Cancer Institute)

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Some of the reference citations in this summary are accompanied by a level-of-evidence designation. These designations are intended to help readers assess the strength of the evidence supporting the use of specific interventions or approaches. The PDQ Supportive and Palliative Care Editorial Board uses a [formal evidence ranking system](#) in developing its level-of-evidence designations.

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